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NTU EMPLOYEE CLAIM SYSTEM · MODULE BUILD PACK

ECS Pre-Trip Travel Request

— Build Specification

A vendor-ready specification of the Pre-Trip Travel Request module: every screen, field, validation rule, program algorithm and configuration switch needed to build it, plus the design decisions that shaped them.

Audience Implementation vendor / OutSystems build team **Status** Prototype-validated, build-ready

Companion Feature Walkthrough artifact (end-user view)

1 The Pre-Trip module

The Pre-Trip module captures a travel request before a trip, derives entitlements and the estimated cost, routes it for approval, issues a Travel Authorisation, hands a governed booking instruction to a Travel Management Company (TMC), and seeds the downstream Travel Expense (TE) claim.

Purpose

The module closes the gap between an employee's **intent to travel** and a **booked, policy-compliant, financially authorised trip**. It replaces email/spreadsheet approvals and retrospective booking with a single governed flow: capture the trip, derive the entitlement and cost, obtain the right financial authority (DOA), and instruct the TMC — **before any spend is committed**. It then pre-populates the post-trip expense claim so the claim reconciles to what was approved.

OUTCOME

Policy enforced before spend

Caps, class entitlement and exceptions are evaluated and routed to the right approver before booking, not discovered at claim time.

OUTCOME

One Travel Authorisation per trip

A dated, amount-bound TA (TA-YYYY-NNNNNN) is the single source of truth the TMC and the claim both reconcile against.

OUTCOME

Clean TMC hand-off & claim

A governed, field-filtered booking instruction per traveller; a pre-filled TE claim whose DOA is auto-granted when it matches the estimate.

Capabilities

Request capture

Individual, **group** (coordinator-raised, multi-traveller) and **guest** (non-employee) requests.

Itinerary & trip types

Round-trip / one-way / multi-city (multi-country), always-on leg builder, personal segments.

Entitlement derivation

Travel class derived from the itinerary (longest-duration leg), per traveller for groups.

Cost estimation

Airfare, accommodation (cap-checked), ODA (rate × eligible days), incidentals; FX conversion; sponsorship.

Policy engine

Common + pre-trip rules → pass / exception / hard stop; hard stops block submission.

Configurable approval routing

RO · Additional · Funding Owner · Cross-BA · Exception · DOA — computed from context + settings.

Charging & allocation

ECS cascade, per-line multi-account splits, auto-allocate by Workday department.

Authorisation & booking

TA issue, validity & booking-deadline tracking, booking-status lifecycle.

TMC hand-off

Governed payload (integration contract), group fan-out (one instruction per traveller), inbound status.

Claim seeding

Pre-populates the TE claim; DOA auto-granted within tolerance.

Amendments & audit

Material-amendment re-approval, immutable version snapshots, field-level audit.

Configuration

Runtime Module Settings drive routing switches, thresholds, caps basis and tolerances.

Scope & boundary

The module **extends the existing ECS / ECS-Travel platform**. It does not re-implement HR, finance or DOA masters — it **reads** them. The build team replaces the prototype's mock service layer with real calls to ECS/OutSystems.

REUSED (READ-ONLY)

ECS reference masters

Employees & appointments, reporting lines, DOA lines, charging accounts (CC/WBS), policy caps & ODA rates, FX rates, workflow amount bands, roles, TMC providers, location master.

NEW (TRANSACTIONAL)

Pre-Trip data store

Travel requests, travellers, itinerary legs, estimated expenses, charging allocations, policy checks, approval steps, bookings, TE claims, audit, attachments.

INTEGRATION SEAM

Mock ECS services

`EcsIdentity`, `EcsCharging`, `EcsWorkflow`,
`EcsRoles`, `EcsReference`, `EcsFx`,
`EcsTravelClassRegister` — the interface the
vendor wires to live ECS.

Stack note. The reference prototype is built on Next.js (server actions), Prisma and SQLite. This spec describes **behaviour, data and logic** — not the prototype stack. On OutSystems, screens map to Screens/Blocks, server actions to Server Actions, the data model to Entities, and the mock services to integration (REST / service actions). Field types and rules below are platform-agnostic.

End-to-end flow

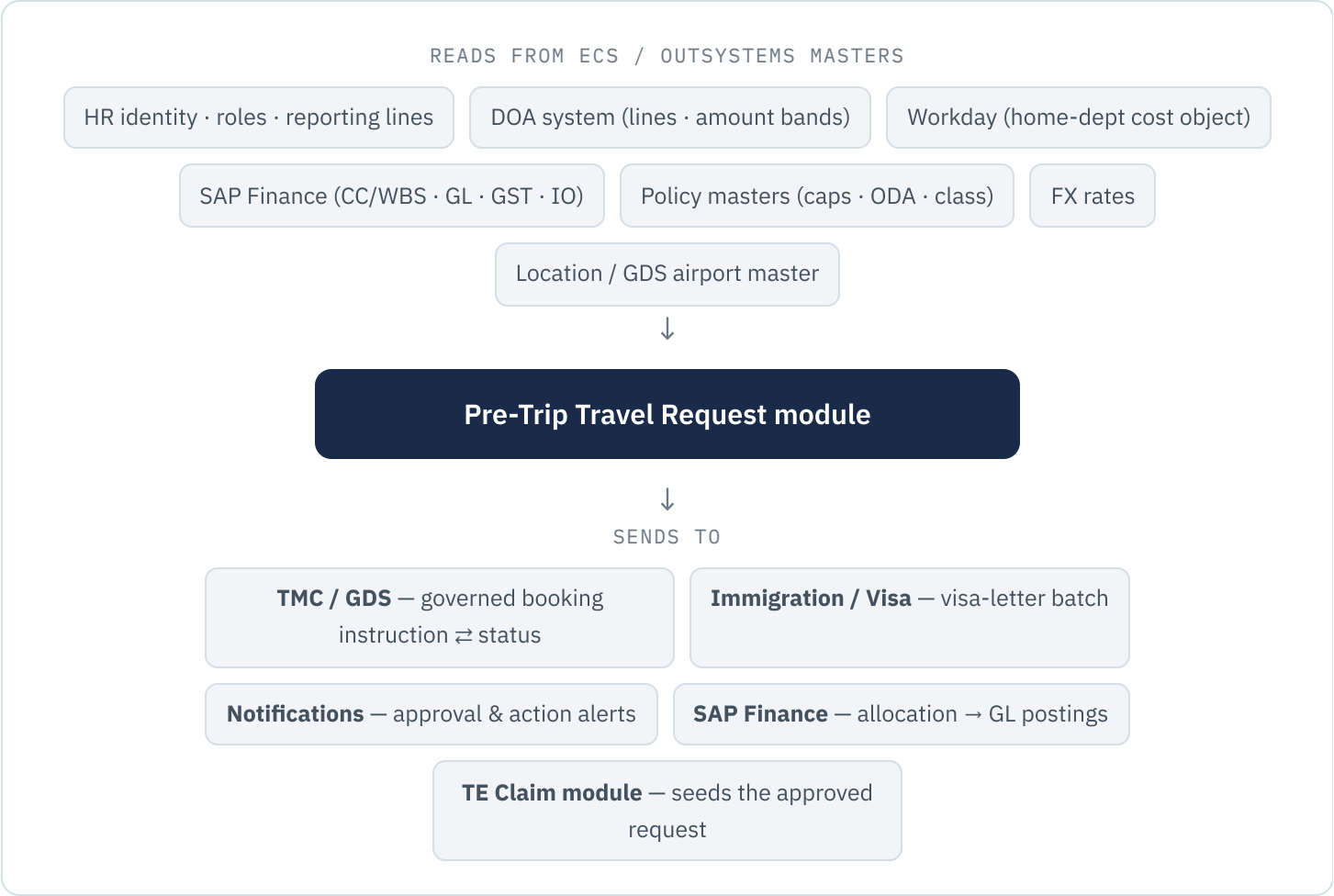
```
# happy path
Create → Trip (purpose, booking, dates, itinerary)
      → Estimates (airfare / accommodation / ODA / other – per traveller)
      → Charging (cost object + allocation)
      → Policy review (caps, class, exceptions)
      → Review & submit (pick DOA, declare)
      → Approval route (R0? ▶ Funding/Cross-BA? ▶ Exception? ▶ DOA)
      → Approved + Travel Authorisation (TA-YYYY-NNNNNN)
      → TMC hand-off (booking instruction) → Booked
      → Trip taken → TE claim (pre-populated; DOA auto-granted within tolerance)
```

High-level architecture

Five layers. The module owns the presentation, domain logic and transactional store; it **reads** reference masters and talks to external systems through a thin integration seam.

LAYER	RESPONSIBILITY
Presentation	Wizard screens (Trip → Estimates → Charging → Policy → Submit), approver/DOA queue, TE claim, configuration. Server-rendered with client interactivity for the itinerary, charging splitter and live validation.
Domain logic	Approval-route engine, class-entitlement derivation, cost/ODA/accommodation computation, policy evaluation, charging roll-up, claim workflow & auto-grant.
Integration seam	ECS service interfaces (<code>EcsIdentity</code> · <code>EcsCharging</code> · <code>EcsWorkflow</code> · <code>EcsRoles</code> · <code>EcsReference</code> · <code>EcsFx</code> · <code>EcsTravelClassRegister</code>) + the TMC adapter and notification/visa hooks. The one place wired to live systems.
Transactional store	Requests, travellers, legs, estimates, allocations, policy checks, approval steps, bookings, claims, versions, audit, attachments.
Reference masters	Read-only ECS masters (identity, DOA lines, charging accounts, policy rates, class register, locations, FX, TMC providers) — never written by this module.

Interaction with other systems



SYSTEM	DIRECTION	WHAT FLOWS
ECS / ECS-Travel core	Read	Employee identity, appointments, roles, reporting lines; the request is created in the ECS context.
DOA system	Read	Per-department DOA lines (+ research line) and amount bands → the eligible DOA candidates and the tier.
Workday (HR)	Read	Each employee's home-department cost object (<code>defaultChargingCode</code>) — drives auto-allocate-by-department.
SAP / Finance	Read + Write*	Charging accounts (CC/WBS), GL accounts, GST codes, Internal Orders (read). The approved allocation drives the GL postings (*via the downstream claim / finance interface).
Policy masters	Read	Hotel caps, ODA daily rates, travel-class register (policy group, min hours, entitled class).
FX service	Read	Entry-date conversion to SGD, re-locked at submission.
Location / GDS airport master	Read (sync)	IATA airports — the same reference the TMC books against; synced from the TMC/GDS feed in production.
TMC	Outbound + Inbound	Governed booking instruction (field-filtered by the Integration Contract; one per traveller for a group) → booking status / traveller-action responses.
Immigration / Visa office	Outbound	Daily visa-letter batch for approved trips flagged as needing a visa letter (§4.13).
Notifications	Outbound	Approval requests, send-backs, booking actions, deadlines.
TE Claim module	Outbound (seed)	The approved request pre-populates the claim; the claim reconciles to the TA and auto-grants the DOA within tolerance.

2 Core design decisions

These decisions are binding — they shaped the data model, screens and logic below. Deviating from one changes multiple downstream behaviours.

DD-01 • Approval routing is configuration-driven

The route is computed, never hard-coded per user.

An evaluator takes the request context (amount basis, charging department, research flag, exceptions, group flag) and emits an ordered list of approval steps. Decision points are toggled by **Module Settings** (the existing ECS Configuration mechanism), not by code.

Why: the same engine serves the Travel Request and the TE claim; policy changes are config, not a release.

DD-02 • DOA is always required and traveller-selected

Every request ends at a DOA; the requestor picks the approver.

The DOA step is never dropped. The system offers the **eligible DOA candidates** (the charging department's DOA line from the amount-band tier upward, excluding travellers) and the requestor chooses — it is **not pre-selected**. Research routes use the research DOA line.

Why: the DOA holds financial authority; choosing (rather than auto-assigning) records intent and supports delegation.

DD-03 • RO is a configurable decision point

Reporting Officer step = ALWAYS · INDIVIDUAL_ONLY · NEVER.

Default `INDIVIDUAL_ONLY` : present for an individual request, dropped for a group (the group coordinator raises it). Resolved as the traveller's Reporting Officer; skipped if the RO is a traveller on the request (self-approval guard).

DD-04 • Travel class = derived entitlement, not a form input

No class picker on the trip form.

Class is derived from the itinerary: each flown leg's duration gives its entitled class and the **highest wins** — so the **longest-duration leg** drives the class (critical for multi-city). It is shown read-only in the cost items, review and detail pages.

Why: removes manual error and upgrade abuse; entitlement is deterministic from the register + itinerary.

DD-05 · Group travel is per-traveller for class & cost; shared for itinerary

One shared itinerary & dates; each traveller at their own entitlement.

Group members share the itinerary and dates (no per-traveller deviation). But **travel class is per traveller** (each at their own entitled class — the coordinator cannot justify an upgrade on another's behalf), and **all costs are per traveller**: airfare, ODA and accommodation. An upgrade for one person is handled by removing them from the group and raising a single request.

DD-06 · Accommodation is never shared – each traveller books, pays and claims their own room

No shared/apportioned hotel line.

Accommodation creates **one line per traveller** (own room, own nights), flowing to that traveller's own TE claim. "Shared — apportion" survives only for genuinely shared incidentals (e.g. a shared taxi under *Other*), where it is displayed apportioned across the group.

DD-07 · Charging allocation is per-expense-line and fully granular

Each cost line splits across one or more cost centres; lines are already per traveller.

ITEM-mode allocation gives every line a split of `{account, %}` rows (must total 100%). The request-level allocation that drives routing is the **roll-up** of net × % over every line × split. An **"Auto-allocate by traveller's Workday department"** preset fills each line from the attributed traveller's home-department cost centre.

DD-08 · TE claim DOA is auto-granted within a configurable tolerance

No re-approval when the claim matches the approved estimate.

The claim route mirrors the request (Verifier → Exception? → RO? → DOA). The DOA step is **auto-granted** from the approved Travel Request when the claim is within the configured tolerance (% and absolute SGD), adds no new expense type, introduces no new policy exception, and keeps the same charging department. No new expense type can be added at the claim.

DD-09 · One always-on itinerary; trip type is a preset, not a mode

Round-trip / One-way / Multi-city reshape the same leg builder.

The leg table is always shown. Trip type presets it: Round-trip = Outbound + mirrored Return, One-way = one leg, Multi-city = user-determined (add / reorder / remove). A round-trip may carry a personal segment. Main destination (for ODA defaults) = longest-stay leg.

DD-10 · Booking arrangement and TMC booking method are separate

What arranges the trip vs how the TMC fulfils it.

Booking arrangement: Auto (route by policy to a TMC) · a specific TMC provider · Self-booked · Host-arranged · No booking. **TMC booking method** (TMC channel: online / agent-assisted / offline) applies only when arranged via a TMC. The resolved TMC is shown read-only on the trip form.

DD-11 · The airport master is the TMC's booking source

One IATA location master, shared.

From/To on every leg read the shared ECS Location master (IATA airports). Each leg's code is carried verbatim into the TMC booking instruction. In production this master is synced from the TMC/GDS airport feed.

DD-12 · A Travel Request has no "Reject"

Dissent = Send Back.

An approver who disagrees **sends the request back** (→ Sent Back → Draft) to be revised and resubmitted. There is no terminal reject on the request. (The TE claim keeps standard claim controls.)

DD-13 · A group may mix employees and guests; its size is capped at the TMC limit

Guest (non-employee) members fly Economy, are host-settled, and never claim; a configurable limit caps the booking.

A group booking can include **guest members** (external speakers / collaborators with no ECS/HR profile) alongside employees. A guest has no entitlement profile so defaults to **Economy**, is **charged to the host (requestor) department**, and is **settled centrally** — guests have no login and **file no TE claim**. The total membership (employees + guests) may not exceed `groupMaxTravellers` (default **10**), mirroring the TMC group-fare limit; the limit is enforced on create and on add-traveller. Full logic in [§14](#).

Why: real group trips routinely carry an invited guest whose cost the host absorbs; the TMC caps a single group fare, so the cap must be a setting, not a constant.

3 Data model

Transactional entities (new). Status/type fields are constrained strings; allowed values are in [§4](#) and inline. Reference masters are in [§13](#).

TravelRequest — header & lifecycle

FIELD	TYPE	NOTES
requestNumber	string, unique	TR-YYYY-NNNNNN
authorisationNo	string, unique?	TA-YYYY-NNNNNN — set on final approval
status / bookingStatus	enum string	Approval + booking state machines (§4)
isGroup	bool	Multi-traveller request
requestorId / travellerId	employee id	Creator may ≠ traveller; GUEST for non-employee
travellerType, guest*	string	EMPLOYEE GUEST (+ guest name/email/org)
additionalApproverId	employee id?	Optional traveller-selected extra approver
selectedDoaId	employee id?	Traveller-chosen DOA (not pre-selected)
purposeId, description	ref / text	Travel purpose (drives research flag)
tripType	enum	ROUND_TRIP ONE_WAY MULTI_CITY
destCountry, destCity	ref	Derived from longest-stay leg on save
startDate, endDate	date	Official travel window
personalStart, personalEnd	date?	Personal extension (excluded from ODA/accommodation)
eventStartDate, eventEndDate, invitationRef	date / text	Event is a segment within the official window
bookingArrangement	enum	AUTO TMC- <id> SELF_BOOKED HOST_ARRANGED NO_BOOKING
bookingMethod	enum?	TMC channel — only when arranged via a TMC
tmcProviderId	ref?	Resolved/selected TMC (null = auto by policy)
travelClassId, entitledClassId, classBasis	ref / text	Request-level class = derived entitlement
chargingMode	enum	MAIN CLAIM ITEM (allocation entry mode)
approvalAmountSgd, bookingDeadline, authorisationExpiry	num / date	Derived approval snapshot (§13.3/§13.4)

FIELD	TYPE	NOTES
visaLetterRequired, highRiskAck, personalAck	bool	Declarations / flags

TravelRequestTraveller

FIELD	NOTES
employeeId	Group member
isRequestor	Coordinator flag
sharePct	Share of NTU-funded cost
entitledClassId / chosenClassId	Per-traveller class (DD-05)
classBasis / classJustification	Derivation / upgrade reason
confirmed, confirmedAt	Inclusion confirmation

ItineraryLeg

FIELD	NOTES
travellerId	null = shared group leg
seq	Order
originCode / destCode	IATA (shared master)
departDate / arriveDate	Within travel window
durationHours	Derived from dest city → drives class
night	Stay at destination
isPersonal	Personal segment (excl. ODA)
entitledClassId / chosenClassId	Per-leg entitlement

EstimatedExpense — one cost line (per traveller)

FIELD	TYPE	NOTES
travellerId	employee id?	Attributed traveller; null = requestor/shared owner
category / expenseTypeId	enum / ref	AIRFARE ACCOMMODATION ODA CONFERENCE OTHER
currency, foreignAmount, sgdAmount	num	Converted at entry-date FX, re-locked at submit
sponsorForeign, sponsorSgd	num	Expected sponsorship (reduces NTU-funded)
estimateBasis	enum	QUOTED CAP LOWER RATE
isShared, shareMap	bool / JSON	Only for genuinely shared incidentals (DD-06)
chargingCode	ref?	Single per-line account (ITEM mode)
allocJson	JSON?	Per-line multi-account split [{code,pct,io}] (DD-07)
originCode, destCode, proposedClassId, fareCeiling	—	Airfare-specific
+ AccommodationEstimate	1:1	city, nights, personalNights, quotedNightly, capNightly, budgetedNightly, exceptionOutcome
+ ODAEstimate	1:1	country, city, arrive, depart, eligibleDays, personalDays, dailyRate

ChargingAllocation — request-level roll-up

FIELD	NOTES
chargingCode, chargingType	CC / WBS id
companyCode, businessArea	SAP org
percent, amountSgd	Share of allocated amount
internalOrder, isMain, isResearch	SAP IO; main account flag; research

Drives DOA routing (highest-share department), cross-BA concurrence and the finance breakdown.

ApprovalStep

FIELD	NOTES
seq, roleType	RO Additional FundingOwner Exception DOA ResearchDOA
approverId	Resolved approver
status, decidedAt, comments	Pending Approved SentBack
autoGranted	TE claim DOA auto-grant (DD-08)

TravelExpenseClaim & TEEexpenseLine

ENTITY · FIELD	NOTES
Claim.claimNumber, claimantId, status	Draft → Submitted → ... ; one claimant
Claim.lessCorpCard, lessPersonal	Deductions from reimbursable total
Claim.chargingMode, chargingMainCode, chargingJson	Allocation snapshot (same editor as the request)
Claim.approvalSteps[]	Verifier → Exception? → RO? → DOA (auto-grant)
Line.expenseTypeId, treatment	LOCKED EDITABLE REFERENCE EXCLUDED (from the TR estimate)
Line.approvedSgd, bookedSgd, actualSgd, varianceSgd	Pre-populated from the TR; variance tracked
Line.receiptType, receiptOk	LOCAL_GST FOREIGN_NO_GST NO_RECEIPT
Line.chargingCode, allocJson	Per-line (multi-)account split, as the request

4 Status model & lifecycle

Two independent state machines on an approved request: **approval status** and **booking status**. Invalid transitions must not be offered.

Approval status transitions

FROM	ALLOWED TO
Draft	Submitted · Withdrawn
Submitted	Pending Additional · Pending Exception · Pending DOA
Pending RO Approval	Pending DOA · Pending Exception · Sent Back
Pending Additional Approval	Pending Exception · Pending DOA · Sent Back
Pending Exception Approval	Pending DOA · Sent Back
Pending DOA Approval	Approved · Sent Back
Approved	Amendment In Progress · Cancelled · Expired · Closed
Amendment In Progress	Approved · Sent Back · Cancelled
Sent Back	Draft · Withdrawn
Expired	Pending RO · Pending DOA · Cancelled (re-validate)

No Reject (DD-12). There is no transition to a terminal "Rejected" on a Travel Request — disagreement routes through **Sent Back** → **Draft**.

Booking status (post-approval, independent)

- Not Sent
- Sent to TMC
- Received by TMC
- Booking In Progress
- Traveller Action Required
- Partially Booked
- Booked
- Travel Completed
- Self-Booked
- Booking Failed
- Cancelled

5 Screens

Each step is a stage in the request wizard (Traveller → Trip → Estimates → Charging → Policy Review → Submit), plus the approver, claim and configuration surfaces. Fields, layout and behaviour below; validation is consolidated in [§6](#).

Entry point. Buckets of the user's requests with a load-demo seed for non-production environments.

Content

- Summary tiles: **Drafts & sent back**, **Pending approval**, **Approved — unbooked**, **Completed — unclaimed**.
- All-requests table: Request · Traveller · Purpose · Destination · Travel dates · Est. NTU cost · Exceptions · Status · Booking.
- Group requests tagged **Group** ; row links to the detail page.

PROTOTYPE UI

NTU

NTU EXPENSE CLAIM SYS...
Pre-Trip Travel Request module

Acting
as

Requestor (traveller) ▾

Dr Alice Tan — Senior Lecturer ▾

AT
Dr Alice
Tan
TRAVELLER

Travel Request Dashboard

+ Create Travel Request

RO / DOA Dashboard

Travel Administration

Finance Oversight

Reporting

Visa Letters

Notifications

Integration Contracts

Configuration Viewer

SETUP & MAINTENANCE

GST Codes

Entertainment Tier

Mileage

CTC Cards

Rejection Reasons

Employee Tier

Define Research WBS

Announcements

TR-01

Travel Request Dashboard

+ Create Travel Request

Drafts, pending actions, upcoming travel, approved-unbooked and completed-unclaimed trips.

0
Drafts & sent back

2
Pending approval

4
Approved —
unbooked

1
Completed —
unclaimed

All Travel Requests (8)

Request	Traveller	Purpose	Destination	Travel dates	Est
TR-2026-000008	Dr Alice Tan	Training/Conference - Acad Staff	London	2026-12-01 → 2026-12-06	\$4,8
TR-2026-000007	Dr Grace Ho	Training/Conference - Acad Staff	Tokyo	2026-11-16 → 2026-11-19	\$4,7
TR-2026-000006	Dr Alice Tan	Training/Conference - Acad Staff	Tokyo	2026-11-02 → 2026-11-10	\$5,1
TR-2026-000005	Dr Alice Tan	Training/Conference - Acad Staff	Tokyo	2026-12-07 → 2026-12-11	\$6,1
TR-2026-000004	Dr Alice Tan	Training/Conference - Acad Staff	Tokyo	2026-11-09 → 2026-11-13	\$9,1
TR-2026-	Dr Henry	Official Business - Research Staff	London	2026-09-20 → 2026-09-27	\$5,7

Dashboard – summary tiles (drafts, pending, approved-unbooked, completed-unclaimed) and the all-requests table.

Request header, approval status, workflow visual, and (for groups) the traveller roster.
Read-only summary of the approved request; the editing wizard steps follow.

Header summary

Purpose · Destination · Dates

Travel class

Booking arrangement · TMC

Primary charging account

Trip header

Individual: entitled class. **Group:** "Each traveller flies at their own class entitlement" + a **Class column in the roster**

Resolved TMC provider shown

Main CC/WBS + %

Group roster (isGroup)

· Traveller · Department · **Class** · Travel dates · Est. cost · Exception · Booking — one row per traveller, each at their own entitled class.

Actions (post-approval)

Booking · Send to TMC · Self-booked · Create TE claim · More actions (amend, cancel).
Workflow chevrons show the realised route with approver names.

PROTOTYPE UI

NTU

NTU EXPENSE CLAIM SYS...
Pre-Trip Travel Request module

Acting as

Requestor (traveller)

Dr Alice Tan — Senior Lecturer

AT

Dr Alice Tan
TRAVELLER

Travel Request Dashboard

Create Travel Request

RO / DOA Dashboard

Travel Administration

Finance Oversight

Reporting

Visa Letters

Notifications

Integration Contracts

Configuration Viewer

SETUP & MAINTENANCE

GST Codes

Entertainment Tier

Mileage

CTC Cards

Rejection Reasons

Employee Tier

Define Research WBS

Announcements

Group Travel Request

GROUP

TR-2026-000004

SUBMITTED BY

Priya Nair (Coordinator)

LAST UPDATED

2026-10-06

Approval Status

Approved

Booking Status

Not Sent

Booking →

Send to TMC

Self-booked

Create TE claim →

More Actions ▾

Workflow

Submitted by Priya Nair

Funding Owner Approved Prof Farah Aziz

DOA Approved Prof Daniel Sim

Approved TA-2026-000003

Travel Purpose

Training/Conference - Acad Staff

Destination

Tokyo, Japan

Official Travel Dates

2026-11-09 – 2026-11-13 (5D/4N)

Proposed Travel Class (per traveller)

Each at own entitlement — see roster

Travel Authorisation Validity

2026-10-06 – 2026-12-13

Event / Conference

Group conference trip

Booking arrangement

Auto — route by policy (TMC) · TMC agent-assisted

Primary Charging Account

CC-1000 · SCSE Teaching & Operations (60%)

Personal Extension

0 travellers

Group Coordinator

Priya Nair

Trip Description / Business Justification

Group conference trip — three travellers (S11)

Group request detail – approval status, realised workflow (Submitted → Funding Owner → DOA → Approved + TA number), and the per-traveller travel-class statement.

Priya Nair

Justification

Group conference trip — three
travellers (S11)

A. Group Summary

3

Total
Travellers

0

Travellers
with
Exception

0

Travellers
with
Personal
Extension

Estimated Total Cost (SGD) **\$9,150.00**

B. Policy Exceptions

● Pass No policy exceptions.

C. Traveller Roster (3)

#	Traveller	Department	Class	Travel Dates	Est. Cost (SGD)	Exception	Bc
1	Dr Alice Tan	School of Computer Science & Engineering	Premium Economy	2026-11-09 – 2026-11-13	\$3,050.00	No	Nc Se
2	Dr Grace Ho	School of Computer Science & Engineering	Economy	2026-11-09 – 2026-11-13	\$3,050.00	No	Nc Se
3	Dr Henry Ong	Research Institute for Artificial Intelligence	Business	2026-11-09 – 2026-11-13	\$3,050.00	No	Nc Se
Total (3 travellers)					\$9,150.00	0	0, Bc

D. Estimated Group Travel Cost Items

Click a row to see how the amount is computed (cap, nights, rate, eligible days).

N

Group summary + Traveller Roster – each member with their department, own entitled class (Premium Economy / Economy / Business) and cost share.

Purpose, booking arrangement, dates and the always-on itinerary. **No travel-class input** (DD-04).

Trip Purpose & Justification

Travel purpose	Reference list; research purposes flag the research route
Booking arrangement	Auto (TMC) · provider · Self-booked · Host-arranged · No booking (DD-10)
TMC booking method	TMC channel — enabled only for TMC arrangements
TMC (resolved)	Read-only — resolved provider from the arrangement (e.g. "CTC Travel (Asia-Pacific)")
Description, Invitation ref, Visa letter	Justification + §4.13 visa flag

Dates — ordered Official → Event → Personal

Official start/end frame the whole trip; the **event** (§10) is a segment within them; **personal** extension (§4.5) extends before or after. The outbound & return legs pre-fill from the official dates.

Destination & Itinerary (DD-09)

- **Trip type** selector presets the leg table (Round-trip 2 · One-way 1 · Multi-city N).
- **Main destination** indicator (read-only) — derived live from the longest-stay leg.
- **Itinerary legs**: Leg · From · To · Depart · Arrive · Nights · Personal (+ add/reorder/remove for multi-city). From/To = IATA master, shown "CODE — Airport name".

PROTOTYPE UI

Integration Services

Configuration Viewer

SETUP & MAINTENANCE

- GST Codes
- Entertainment Tier
- Mileage
- CTC Cards
- Rejection Reasons
- Employee Tier
- Define Research WBS
- Announcements

CTC Travel (Asia-Pacific) · derived from the booking arrangement on save (\$2.2)

Description / justification

Group conference trip — three travellers (\$11)

Invitation / acceptance reference

e.g. IEEE-2027-ACCEPT-4821

☐ A **visa letter** is required for this trip (\$4.13) — the immigration office is notified once the request is approved.

Dates

Official start

09/11/2026

Official end

13/11/2026

Event start (\$10)

dd/mm/yyyy

Event end

dd/mm/yyyy

Personal extension start

dd/mm/yyyy

Personal extension end

dd/mm/yyyy

Official dates frame the whole trip; the event (\$10) is a segment within them and personal days (\$4.5, AC18) extend before or after — personal days are excluded from ODA and personal nights from the accommodation budget. Set the official dates first — the outbound and return legs below pre-fill from them.

Destination & Itinerary

Trip type

Round-trip

One-way

Multi-city

There and back to one destination — two legs (a personal segment is allowed).

MAIN DESTINATION

Tokyo, Japan · derived from the longest-stay leg (HND)

Itinerary legs

Leg	From	To	Depart	Arrive	Nights	F
Outbound			09/11/2026	dd/mm/yyyy		

Trip form – **resolved TMC** (read-only), Dates ordered Official → Event → Personal, the trip-type preset, the derived **Main destination** and the always-on itinerary legs. No travel-class input (DD-04).

Per-traveller cost lines, pre-filled from the trip. Four add-forms: Airfare, Accommodation, ODA, Other/Conference.

List (group = collapsible, one row per expense type)

Grouped by cost item with a subtotal; expand to the per-traveller rows. Individual = flat table. Gross estimate → less sponsorship → NTU-funded.

Add-forms – group behaviour

Traveller selector

Individual traveller **or "All travellers"** – one line per traveller in a single submit

Airfare · Proposed class

Group: **read-only** "per traveller – set to entitlement"; line stamped with each traveller's class

Accommodation

Per traveller (own room) – no Shared checkbox (DD-06)

Other / Conference

May be **Shared – apportioned** for genuinely shared incidentals

ODA is auto-computed from policy rate × eligible days over the official range (personal days excluded).
Accommodation shows quoted/night vs cap and flags an exception over the cap.

PROTOTYPE UI

NTU

NTU EXPENSE CLAIM SYS...

Acting as

Requestor (traveller)

Dr Alice Tan — Senior Lecturer

AT

Dr Alice Tan

TRAVELLER

Travel Request Dashboard

Create Travel Request

RO / DOA Dashboard

Travel Administration

Finance Oversight

Reporting

Visa Letters

Notifications

Integration Contracts

Configuration Viewer

SETUP & MAINTENANCE

GST Codes

Entertainment Tier

Mileage

CTC Cards

Rejection Reasons

Employee Tier

Define Research WBS

Announcements

Traveller

Trip

Estimates

Charging

Policy Review

Submit

Estimated Travel Costs (9)

Airfare · 3 traveller lines				\$3,750.00
Overseas Daily Allowance · 3 traveller lines				\$1,440.00
Overseas Accommodation · 3 traveller lines				\$3,960.00
Dr Alice Tan	Tokyo · 4n · quoted \$330.00/n vs cap \$350.00/n	\$1,320.00	Pass	Remove
Dr Grace Ho	Tokyo · 4n · quoted \$330.00/n vs cap \$350.00/n	\$1,320.00	Pass	Remove
Dr Henry Ong	Tokyo · 4n · quoted \$330.00/n vs cap \$350.00/n	\$1,320.00	Pass	Remove
Gross estimate				\$9,150.00

Add Airfare (TR-06)

Origin

SIN

Destination

HND

Proposed class

Per traveller — set to entitlement

Expected date

09/11/2026

Currency

SGD

Quoted amount

Add Accommodation (TR-07)

City

Tokyo

Quoted rate / night

Nights

4

Personal nights

0

Check-in

09/11/2026

Check-out

13/11/2026

Conference hotel

Group estimates – collapsible, one row per cost item; Overseas Accommodation expanded to **one room per traveller** (DD-06). Add-forms show the per-traveller class and the Traveller / All-travellers selector.

ECS 4-level cascade + allocation. Drives DOA routing.

Estimated Cost Lines

Type · **Traveller** (group) · Detail · Gross · Sponsor · NTU-funded, with the NTU-funded total.

Main charging account – cascade

Company Code → Business Area → CC/WBS → Charging Account. Defaults to the traveller's profile cost object. Closed accounts cannot be charged.

Cost allocation modes

- **MAIN** — whole request to the main account.
- **CLAIM** (Allocation at Claim/Item Level) — a %/amount split of the whole request across accounts.
- **ITEM** (Allocation per Expense Item) — **per-line splitter** grouped by cost item; each line splits across 1+ accounts (% to 100). **Auto by dept** per line + **Auto-allocate all by traveller's department** (DD-07).

Allocation Outcome (TR) / Accounting Entries (TE)

Computed GL postings per account × expense type: Company · BA · CC/WBS · Internal Order · GL · Expense type · Amount · % · Main acct · GST. Must total 100% before submit.

PROTOTYPE UI

(CC-2000) RI-AI Operation

100

Int. order

SGD 480

+ split

Auto by dept

Overseas Accommodation

SGD 3,960

Dr Alice Tan

SGD 1,320

(CC-1000) SCSE Teaching &

100

Int. order

SGD 1,320

+ split

Auto by dept

Dr Grace Ho

SGD 1,320

(CC-1000) SCSE Teaching &

100

Int. order

SGD 1,320

+ split

Auto by dept

Dr Henry Ong

SGD 1,320

(CC-2000) RI-AI Operation

100

Int. order

SGD 1,320

+ split

Auto by dept

Each line's split must total 100%. The Allocation Outcome below rolls every line's split up to the request-level charging accounts.

Allocation Outcome

Company	Business area	CC/WBS charging account	Internal Order	GL account	Expense type	Amount	alloc
1000	BA-CS	CC-1000	—	GL-6120	Overseas Accommodation	SGD 2,640	6i
1000	BA-CS	CC-1000	—	GL-6110	Airfare	SGD 2,500	6i
1000	BA-CS	CC-1000	—	GL-6130	Overseas Daily Allowance	SGD 960	6i

ITEM-mode allocation auto-allocated by department: each per-traveller room routes 100% to that traveller's dept (Alice/Grace → CC-1000, Henry → CC-2000), and the **Allocation Outcome** rolls up the GL postings.

5.6 Policy review

/requests/{id}/policy

Runs the common + pre-trip rules and lists outcomes: **Pass** **Exception** **Hard stop**. Hard stops block submission; exceptions are carried to the Exception Approver or surfaced to the DOA per config.

Approval Route (visual) – above the Trip card

Chevron preview of the computed route. The requestor **selects the DOA** (required; from eligible candidates; not pre-selected) and may add one Additional Approver – the chevrons update live.

Trip card

Purpose · Destination · Dates · Travel class (group = per-traveller statement) · Booking arrangement · **TMC** · Traveller. Multi-city shows the full leg list.

Estimated cost items

Grouped by type (group = expandable to per traveller): Gross → less sponsorship → Estimated NTU-funded → Approval amount (per basis) → approved booking ceiling.

Declaration & submit

Official-purpose + conflict-of-interest declaration; high-risk acknowledgement when applicable. Submit disabled while any hard stop remains or no DOA is chosen.

PROTOTYPE UI

NTU

NTU EXPENSE CLAIM SYS...

Acting as

Requestor (traveller)

Dr Alice Tan — Senior Lecturer

AT

Dr Alice Tan

TRAVELLER

Travel Request Dashboard

Create Travel Request

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SETUP & MAINTENANCE

GST Codes

Entertainment Tier

Mileage

CTC Cards

Rejection Reasons

Employee Tier

Define Research WBS

Announcements

Traveller

Trip

Estimates

Charging

Policy Review

Submit

Approval Route

Traveller submits

Funding Owner / Cross-BA
Prof Farah Aziz

DOA
Select the DOA

Approving DOA *

Additional Approver (optional)

— select the DOA —

— none —

Pulled from the DOA system for the charging department & amount band (\$13.14); choose who approves — it is not pre-selected.

Optionally route through one additional approver before the DOA (\$23). You cannot select yourself.

Trip

PURPOSE

Training/Conference - Acad Staff

DATES

2026-11-09 → 2026-11-13

TRAVEL CLASS

Each traveller flies at their own class entitlement (\$13.14) — per-traveller classes are shown in the cost items below.

BOOKING ARRANGEMENT

Auto — route by policy (TMC) · TMC agent-assisted

TRAVELLER

Dr Alice Tan

DESTINATION

Tokyo, Japan

TMC

CTC Travel (Asia-Pacific)

Charging

CC-1000	Non-research	60%
CC-2000	Non-research	40%

Review & submit – the computed **Approval Route** chevrons (Traveller → Funding Owner / Cross-BA → DOA), the required **DOA selector** (not pre-selected), and the Trip card with the resolved TMC and per-traveller class statement.

5.8 Approver / DOA dashboard

/approvals

Per-role queue (acting-as role selector). Each pending step shows the request, the approval amount, exceptions, and the right-aligned numeric data. Decisions: **Approve** or **Send Back** (with reason). A high-risk request requires an approver acknowledgement.

5.9 TE claim (Create from TR)

/claims/{id}

Pre-populated from the approved request so the claimant does not re-key. Header (report type, travel dates, deductions), expense lines (pre-filled, treatment-controlled), and the same charging editor.

Pre-population

Each estimate line seeds a claim line (approved amount, charging), with per-line treatment LOCKED / EDITABLE / REFERENCE / EXCLUDED. **No new expense type** may be added at the claim.

Workflow

Verifier → Exception? → RO? → **DOA (auto-granted within tolerance)** — see [§10](#).

5.10 Configuration / Module Settings

/settings · /config-viewer

The existing ECS Configuration mechanism — a single settings row drives routing switches, thresholds, caps basis, declarations and tolerances. Full catalogue in [§11](#).

6 Validation rules

Every rule is enforced **server-side** (authoritative); the client mirrors them for instant feedback and disables the action while a blocking rule fails.

Trip & dates

RULE	ENFORCEMENT
Travel purpose, ≥ 1 itinerary leg with a destination, travel dates, booking arrangement required	Block save
Official end $\geq$ official start	Block
Each leg: arrive $\geq$ depart	Block
Legs chronological — leg $n+1$ depart $\geq$ leg n depart	Block
Each leg departs within the travel window (widened by a personal extension on either side)	Block
Personal extension: end $\geq$ start, and the window does not overlap the official dates (before start or after end only)	Block
Event dates within the official window (§10) — event is a segment of the trip	Block

Charging

- Each cost line's split must total **100%** (ITEM mode); the request-level allocation must total 100% before submit.
- A closed/inactive account cannot be charged.
- Auto-by-department falls back to the main account and flags any traveller with no mapped department cost centre.

Submission

- All **hard stops** resolved (AC06).
- A **DOA is selected** when candidates exist (DD-02).
- High-risk destination acknowledgement checked when applicable (§4.8).
- Status transition is in the allowed set (§4); self-approval prevented (an approver who is a traveller is skipped/escalated).

TE claim

- No expense type absent from the approved request.
- A line claimed with `NO_RECEIPT` > 0 raises a new exception (blocks auto-grant).
- Claim over the approved amount beyond tolerance raises a new exception.

7 Program logic — approval routing

The route builder consumes the request + settings and emits ordered steps. Order:

Additional → RO → Funding Owner → Cross-BA concurrence → Exception → DOA.

```
function buildRoute(req, settings):
  steps = []; travellers = req.isGroup ? req.travellers : [req.travellerId]
  dept = highestChargingDept(req)           # department bearing the largest %
  research = isResearch(req) and not settings.sameRouteResearch

  # 1 · Additional Approver (traveller-chosen, optional) – self-approval guard
  if req.additionalApproverId and notTraveller:
    steps += ADDITIONAL(req.additionalApproverId)

  # 2 · RO decision point (DD-03)
  roNeeded = settings.roRequirement == "ALWAYS"
             or (settings.roRequirement == "INDIVIDUAL_ONLY" and not req.isGroup)
  if roNeeded and R0(traveller) notTraveller: steps += R0

  # 3 · Funding Owner (cross-charge requiring owner concurrence)
  if anyAllocation.crossCharge and fundingOwner: steps += FUNDING_OWNER

  # 4 · Cross-business-area concurrence (§18)
  for each non-primary BA whose share > settings.crossBaThresholdSgd:
    steps += FUNDING_OWNER(thatBA)           # skip duplicates / travellers

  # 5 · Exception approver – only if a policy exception exists AND configured on
  if hasException and settings.exceptionApproverRequired: steps += EXCEPTION

  # 6 · DOA (always) – tier from the amount band, line per research route (DD-02)
  band = bandForAmount(approvalAmount)
  doaId = req.selectedDoaId # traveller-chosen; else pickDoa(line, band.tier, escalate pas
  steps += (research ? RESEARCH_DOA : DOA)(doaId)
  return steps
```

DOA candidate derivation (what the requestor chooses from)

```
function doaCandidatesFor(req, approvalAmount):
  line = research ? dept.researchDoaLine : dept.doaLine
  band = bandForAmount(approvalAmount)
  tier = min(band.doaTierIndex, line.length-1)
  return line.slice(tier)           # required tier upward (higher authority can al
    .filter(id not in travellers)    # never a traveller (self-approval, $13.5)
```

Status derivation. The request's Pending ★ status is set from the first pending step's role (RO → Pending RO Approval, Additional → Pending Additional, Exception → Pending Exception, else Pending DOA).

8 Program logic — travel class entitlement

Entitlement is derived from the **Travel Class Register** (per-employee policy group, entitled class, minimum flight-hours threshold, effective dates) across the whole itinerary.

```
function entitledForDuration(employee, hours, onDate):
  entry = register(employee, onDate)
  return entry and hours >= entry.minHours ? entry.entitledClassId : ECONOMY

function entitledForItinerary(employee, legs, onDate):  # DD-04
  perLeg = [ entitledForDuration(employee, leg.durationHours, onDate)
    for leg in legs if not leg.isPersonal ]
  return highest-rank(perLeg)           # longest-duration leg drives the class
```

- **Individual:** request-level `travelClassId = entitledForItinerary(traveller, legs)`.
- **Group (DD-05):** compute per traveller and store on each `TravelRequestTraveller.chosenClassId / entitledClassId`. Each airfare line is stamped with that traveller's class; lines re-sync if the itinerary changes.
- Leg duration is derived server-side from the destination city's flight-hours — not trusted from the form.

9 Program logic — charging & allocation

Per-line split → request-level roll-up (ITEM mode)

```
function resolveItem(lines):
  charges = []
  for line in lines:
    allocs = line.allocJson or [{code: line.mainAccount, pct: 100}]
    line.save(allocJson = allocs.length>1 ? allocs : null)  # persist genuine splits
    for a in allocs: charges += {code: a.code, net: line.net * a.pct/100}
  rows = rollupByCode(charges)  # sum net per account → % of total
  persist ChargingAllocation(rows)  # drives routing – unchanged contract
```

Auto-allocate by traveller's Workday department (DD-07)

```
function deptAllocsFor(line):
  shares = line.isShared
    ? (line.shareMap or equalSplit(req.travellers))  # shared incidental
    : [{ eid: line.travellerId, pct: 100 }]  # per-traveller line
  byCode = {}
  for s in shares:
    cc = employee(s.eid).defaultChargingCode  # Workday home-dept cost object
    byCode[cc] += s.pct  # aggregate same-dept traveller
  return byCode  # → one {account, %} row per department, editable after fill
```

Because accommodation (and airfare, ODA) are now **per traveller** (DD-06), each line auto-allocates 100% to that traveller's department — **no split**. A shared incidental splits across departments by share. The highest-share department derives the DOA (§13.14).

10 Program logic — TE claim & DOA auto-grant

The claim route reuses the same configurable rules; the DOA step is auto-granted when the claim matches the approved request.

```
function doaAutoGrant(claim, req, settings):  # DD-08
    if req.status not in [Approved, Closed]: return false  "not approved"
    if not req.authorisationNo:               return false  "no TA"
    over = claimTotal(claim) - approvedAmount(req)
    withinTol = over <= settings.teAutoGrantToleranceAbsSgd
               and over <= approvedAmount * settings.teAutoGrantTolerancePct/100
    if not withinTol:                         return false  "beyond tolerance"
    if claimHasNewExpenseType(claim, req):    return false  "new expense type"
    if claimHasNewException(claim, req):      return false  "new exception"
    if claimChargingDept != reqChargingDept:  return false  "DOA changed"
    return true  # DOA step marked Approved · autoGranted, cites the TA
```

New exception = over the approved amount beyond tolerance **or** any line claimed without a receipt. The Verifier (Travel Administrator) is always first; the Exception and RO steps follow the same config as the request.

11 Configuration reference — Module Settings

Single settings row (the ECS Configuration mechanism). Runtime-toggable; changes affect routing and derivation immediately.

SETTING	VALUES · DEFAULT	EFFECT
roRequirement	ALWAYS · INDIVIDUAL_ONLY · NEVER	RO decision point (DD-03)
groupTravelEnabled	bool · true	Enables one-request-many-travellers group requests (DD-05)
groupMaxTravellers	int · 10	Group booking size limit incl. guests — TMC group-fare cap (DD-13)
exceptionApproverRequired	bool · true	Insert an Exception step, else surface to DOA
sameRouteResearch	bool · false	Treat research like non-research (no Research DOA line)
approvalAmountBasis	GROSS · NET_NTU · MAJOR_COST · ALLOCATION	Which amount drives the band / DOA tier
crossBaThresholdSgd	int · 500	Cross-BA concurrence threshold (§18)
teAutoGrantTolerancePct	int · 10	Claim DOA auto-grant tolerance (% over)
teAutoGrantToleranceAbsSgd	int · 500	Claim DOA auto-grant tolerance (abs SGD)
hotelEstimateBasis	QUOTED · CAP · LOWER	Accommodation budget basis
expenseScope	MAJOR_ONLY · ALL	Whether incidental/other lines are captured
chargingSplitMode	PERCENT · AMOUNT	Allocation entry mode
selfApprovalLimitSgd	int · 0	Amount at/below which a traveller-approver self-approves (0 = never)
authorisationValidityDays / bookingDeadlineDays	int · 30 / 14	TA validity & booking deadline (§13.3)
attachmentMaxMb	int · 7	Max upload size per file
declarationText / coiText	text	Submission declarations

12 TMC integration

On approval (and hand-off), a **governed booking instruction** is assembled from the request, filtered by the admin Integration Contract (per-field include/treatment flags).

Hand-off payload (per instruction)

- **Authorisation** & traveller (or guest) identity; booking method; booking deadline; approval validity.
- **Air segments** — origin / destination (IATA, verbatim), depart date, approved class.
Hotel segments — city, nights, capped nightly.
- Approved cost ceiling; approved exceptions; charging (primary bears the airfare posting); approval metadata.
- **Group fan-out (§9.3)**: one approval → **one instruction per traveller**, each carrying that traveller's own class, nights and cost share.

Booking arrangement decides the channel: AUTO routes to a TMC by policy/destination; a provider id targets a specific TMC; Self-booked / Host-arranged / No booking skip the hand-off. The resolved TMC is shown read-only on the trip form and the review/detail pages.

13 Reference masters (read from ECS)

IDENTITY

Employees & appointments

Department, business area, reporting officer, travel policy group, **defaultChargingCode** (Workday home-dept cost object), roles.

AUTHORITY

DOA lines & workflow bands

Per-department DOA line (+ research line); amount bands → DOA tier index + label.

FINANCE

Charging accounts

CC/WBS with company code, business area, department, research flag, funding owner, active/closed.

POLICY

Caps & ODA rates

Per-city hotel caps; per-country ODA daily rates; travel-class register (policy group, min hours, entitled class, effective dates).

LOCATION

Countries · cities · airports

IATA airport master (shared with the TMC); city → country; flight-hours from base.

RATES

FX & expense types

Entry-date FX (re-locked at submit); expense types with GL account & GST code.

VENDORS

TMC providers

Provider + region + booking methods (channels); policy routing.

14 Group & guest bookings

Group travel (DD-05) lets one request carry many travellers on a shared itinerary, each at their own class and cost. A group may also include **guest members** — non-employees with no ECS/HR profile (DD-13). Membership is capped at `groupMaxTravellers` (default **10**), the TMC group-fare limit.

Member model

Each member is a `TravelRequestTraveller` row. A guest is distinguished by `travellerType = GUEST` and a synthetic `employeeId` (a `G-...` key, since there is no HR id); its identity is captured inline.

FIELD	EMPLOYEE MEMBER	GUEST MEMBER
employeeId	ECS/HR id	synthetic G-... key
travellerType	EMPLOYEE	GUEST
guestName / guestEmail / guestOrg	—	captured on the create form
entitled / chosen class	derived per itinerary (DD-04)	Economy (no entitlement profile)
charging	traveller's Workday department	host (requestor) department
confirmation	each confirms inclusion (§13.13)	host confirms offline (pre-confirmed)
TE claim	files own claim	none — settled centrally, host-absorbed

Creation & size limit

```
function createGroupDraft(employeeIds, guests, settings):  # DD-13
  total = employeeIds.length + guests.length
  if total < 2: reject "a group needs ≥2 members"
  if total > settings.groupMaxTravellers: reject "over the TMC group limit"
  for eid in employeeIds: assert canCreateFor(requestor, eid)  # §13.13 delegation
  members = employeeIds.map(EMPLOYEE) + guests.map(g => GUEST("G-"+uid(), g))
  share = 100 / total  # equal share; roll-up drives routing
  create TravelRequest(isGroup, travellers = members)
```

The create form offers the full traveller pool plus inline guest rows (name / email / organisation), shows a live **"N of max M"** counter, and blocks submit when over the limit or under two members. **Add-traveller** on an existing group re-checks the same cap.

Downstream handling

- **Entitlement (§8):** a guest short-circuits to Economy; employees derive per itinerary. Each member's class is stamped on their own airfare line.
- **Charging (§9):** a guest's share maps to the host department's cost object; the per-department roll-up still derives the DOA from the highest-share department. Guest cost is never split onto the guest.
- **Review & detail:** the roster shows each member with a **Guest** pill, their org (*Guest* · *MIT*), and Economy class.
- **TMC fan-out (§12):** a guest instruction carries `bookingFor = GUEST`, the guest identity block, and Economy class; employee instructions are unchanged.
- **Claim:** claims are keyed to the logged-in persona, so a guest — having no login — can never produce one; the host absorbs guest cost.

A per-person upgrade or a guest who must claim their own cost is handled by **removing them from the group and raising a single request** (the DD-05 escape hatch), not by special-casing the group.

ECS Pre-Trip Travel Request — Build Specification · NTU Employee Claim System.
Companion to the end-user Feature Walkthrough. Section references (§) map to the Build Pack. This document describes behaviour, data and logic; the implementation platform (OutSystems) and the prototype stack (Next.js/Prisma) are separate concerns.